FOUNDATION FOR
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RESEARCH

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	22-000415
Award Program	ROAR
Project Title	Efficacy of Live-vectored Prototype ASFV Subunit Vaccines
Grant Start Date	08/15/2022
Grant End Date	06/30/2023
Total Grant Budget	$300,000.00
Total FFAR Budget	$150,000.00
Matching Funder(s)	MEDIAN Diagnostics Inc.
Final Report Date	10/25/2023
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General Information

1.1. Please list the geographic location(s) – city, state, congressional district - wherethe work was conducted. If the work was conducted outside of the US, please list the city and country.

Manhattan, Kansas, Congressional district: 1st.

1.2. How many new jobs were created because of this grant funding? 9

1.3. How many jobs were maintained because of this grant funding? 2

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profitstatus since you were awarded the grant? If yes, please explain. No.

1.5. Has your organization undergone a recent name change? If so, please providethe new name of your organization. No.

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

This project is addressing the danger posed by the African Swine Fever Virus (ASFV), which is a High Priority National Food Security threat, but there is no vaccine or treatment available. The only management option is culling affected swine populations and implementation of strict biosecurity. The virus has spread to many parts of Europe, Russia, Asia, and more recently the Dominican Republic and Haiti, thus posing imminent danger to the U.S. swine industry. Notably, the U.S. swine agriculture is a multi-billion-dollar industry that is a major source of food, revenue, and employment as well as a customer for goods and services. Presence of feral pigs across many states and ticks capable of harboring the virus means that an outbreak could result in establishment of a domestic virus reservoir that could be difficult to eliminate as it happened in wild boars in Europe and Russia. An ASFV

outbreak will cost billions of dollars and jeopardize food security and therefore, development of an effective, affordable, and safe vaccine for long-term ASFV control is a high priority. Development of safe vaccines is feasible since immunization of pigs with some naturally attenuated or engineered ASFV can induce protection, which imply that the virus has proteins that induce protection. However, attenuated ASFV vaccines are yet to be commercialized, due to safety and production limitations. More importantly, the ASFV proteins that induce protection are yet to be identified and an effective method for immunization of pigs is lacking. In this project, an experimental vaccine containing a cocktail of ASFV proteins was generated and validated in readiness for testing in pigs. The vaccine was safe and well tolerated following immunization of pigs with three doses. Antibodies induced by the vaccine strongly recognized the actual ASFV. However, following exposure of the pigs to ASFV, only one pig survived until the end of the study, but it failed to completely clear the virus even though it had high levels of antibodies against ASFV. Subsequent studies showed that the antibodies induced by the ASFV proteins did not neutralize the virus or block infection of swine cells by the virus. On a positive note, screening of peptide pools from the ASFV proteins using cells from immunized pigs resulted in identification of seven proteins with potential to induce protective T cell responses. Fine mapping resulted in identification of two peptides (epitopes) that stimulated T cells from immunized pigs to secrete a molecule (Granzyme B) that is involved in killing virus-infected cells. These two peptides are 100% conserved among all the sequenced ASFV genomes and they have potential to induce T cell responses in pigs from different genetic backgrounds. Such highly conserved epitopes will be excellent candidates for development of broadly protective subunit vaccines. The identification of seven proteins and the two highly conserved epitopes with potential to play a role in inducing protection in pigs is new knowledge that will guide development of a safe vaccine that is needed to safeguard the U.S. pork industry.

2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Given that some attenuated ASFV isolates can induce protection, it is rational to determine the swine immune response mechanisms required for protection and more importantly, empirically identify protective antigens from the whole proteome to inform rational design and development of a prototype subunit vaccine capable of safely conferring sterile immunity. This was a proof-of-concept study in which one prototype adenovirus-vectored multiantigen ASFV subunit vaccine was generated and validated *in vitro* for protein expression and authenticity by using well characterized convalescent serum. The experimental vaccine was safe and it induced strong antibody responses in domestic pigs (n=5), but only conferred protection to one vaccinee. The antibodies induced by the adenovirus-vectored antigen cocktail strongly recognized the actual ASFV, but failed to neutralize the virus or block infection of swine cells *in vitro*. This finding suggests that the antibodies directed against the antigens included in the cocktail are not protective. This finding, if confirmed, will be critical for guiding selection of antigens for inclusion in the development of a protective subunit vaccine.

Notably, seven novel ASFV antigens [A118R, 173R, M1249L, G1211R, A859L, B318L, and B169L] that recalled strong ASFV-specific granzyme-B^+ and IFN-γ^+ responses by $CD8\alpha^+$ T cells following *in vitro* re-stimulation of peripheral blood mononuclear cells (PBMCs) by ASFV (Georgia 2007/1) were identified. Fine mapping resulted in identification of potent granzyme-B^+IFN-$\gamma^+CD8\alpha^+$ T cell epitopes from antigen A859L: A859L156-164 (STDETRTAI); and A859L428-436 (LVDQGVYAL). These peptides are 100% conserved among all the sequenced ASFV genomes and they are restricted by multiple *SLA-I* alleles [*Zajac, M., et al., 2023: manuscript in-preparation*]. Such highly conserved epitopes will be excellent candidates for development of broadly protective subunit vaccines. Recent findings suggest that cytotoxic T lymphocytes (CTLs) capable of killing ASFV-infected cells play a critical role in virus clearance. Normally, lytic activity of CTLs involve release of granzyme-B^+ effector molecule and therefore, if confirmed, the above-mentioned results suggest that the seven antigens capable of inducing ASFV-specific granzyme-B^+IFN-$\gamma^+CD8\alpha^+$ T cell responses will be potential subunit vaccine candidates. This new knowledge will guide selection of antigens for inclusion in a prototype subunit vaccine.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and classified in one of the following ways:

a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

The ultimate project goal is to develop an efficacious subunit vaccine capable of safely inducing sterile immunity in pigs. To achieve this goal, and backed by the fact that attenuated ASFV can confer protection, it is critical for protective ASFV antigens to be identified to allow rational vaccine design and optimization of a safe and efficacious prototype subunit vaccine. Accordingly, execution of this project resulted in the demonstration that replication-defective adenovirus-5 (RD-Ad5) encoding multiple ASFV antigens can elicit strong antibody responses that recognized the virus, but *in vitro* analyses showed that the antibodies induced by the antigens included in the cocktail failed to neutralize the virus or block infection of primary swine cells. This finding will be important to us and to other researchers with regards to antigen selection, design and development of a subunit ASFV vaccine. Following priming and two homologous boosts, analyses of cellular immune responses showed that the experimental vaccine induced ASFV-specific memory T cells that were recalled, *in vitro*, upon stimulation with low dose live ASFV. Screening ASFV antigens using the recalled memory T cells resulted in identification of seven antigens that stimulated significant secretion of granzyme-B^+ and IFN-γ^+ by CD8α^+ T cells, an outcome that is consistent with the current thinking regarding the critical effectors required for lysis of ASFV-infected cells by CTLs in pigs. The above outcomes showed that immunization of pigs with the RD-Ad5 vectored experimental vaccine induced high antibody responses that were not protective, but low level of granzyme-B^+IFN-γ^+CD8α^+ memory T cells responses that could be involved in virus clearance. The low level of memory T cells observed after immunization with three doses suggest that the RD-Ad5 vector might not be effective at priming and expanding sufficient T cell responses required to eliminate ASFV-infected cells. This could be due, in part, to the inability of the recombinant RD-Ad5 to amplify *in vivo*. This outcome prompted us to develop a replication-competent live-vector that can replicate in pigs and thereby prime and significantly expand effector and memory T cells in a manner similar to attenuated ASFV which replicates *in vivo*. The outcome will also guide other investigators working on ASFV subunit vaccine development.

The identified seven ASFV antigens [A118R, 173R, M1249L, G1211R, A859L, B318L, B169L] that recalled strong ASFV-specific granzyme-B^+IFN-γ^+CD8α^+ T cell responses is new knowledge that will be considered in development of the second-generation prototype subunit vaccine to be evaluated in pigs. In addition, the A859L antigen was shown to contain two epitopes (STDETRTAI; and LVDQGVYAL) that induced strong granzyme-B^+IFN-γ^+CD8α^+ T cell responses. Importantly, the epitopes are 100% conserved among all the sequenced ASFV genomes and they are restricted by multiple *SLA-I* alleles. Therefore, the A859 antigen is an attractive candidate to be considered for development of a broadly protective ASFV subunit vaccine. The above findings will certainly require follow up studies, preferably using a replication-competent vector, to confirm suitability of the antigens for inclusion in a next generation prototype subunit vaccine to be evaluated in pigs. Given that ASFV is a large and complex virus with more than 160 antigens, comprehensive whole proteome empirical identification of vaccine candidate antigens is the only path that will lead to generation of new knowledge needed to guide rational design of an effective subunit vaccine.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The overall project goal is to develop a safe and efficacious African Swine Fever Virus subunit vaccine. The specific objectives were:

1) Empirically identify and validate the actual ASFV antigens that stimulate protective immunity. The first part entailed homologous prime-boost immunization of pigs with either Adenovirus 4 (Ad4)- or Adenovirus 6 (Ad6)-vectored prototype subunit vaccines, designated Ad4-PV and Ad6-PV, encoding

multiple ASFV antigens selected based on their potential to induce porcine granzyme B^+ T cells. The second part was focused on heterologous priming with Ad4-PV and boosting with Ad6-PV, or Ad6-PV prime and Ad4-PV boost. Sera and lymphocytes from survivors would be used to identity immune responses that correlate with protection as judged by comparison to responses or lack thereof in non-survivors. See section 2.10.

2) Formulate an optimized prototype vaccine that will be evaluated in clinical trials.

2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit). No.

2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

The ultimate project goal is to develop a safe and effective ASFV subunit vaccine, which is within reach since attenuated ASFV can confer protection. To achieve this goal, it is imperative to empirically identify protective ASFV antigens since none has been defined so far and equally important, find a suitable vector capable of effectively inducing immune responses capable of clearing virus in swine. Two replicon vectors, Adenovirus 4 (Ad4) and Adenovirus 6 (Ad6) were explored to generate prototype subunit vaccines, designated Ad4-PV and Ad6-PV, encoding multiple ASFV antigens selected based on their potential to induce porcine granzyme B^+ T cells. The objectives were to determine immunogenicity and protective efficacy of the prototype vaccines following: i) homologous prime-boost immunization of pigs with either Ad4-PV or Ad6-PV; or ii) heterologous priming with Ad4-PV and boosting with Ad6-PV, or Ad6-PV prime and Ad4-PV boost. Following challenge, sera and lymphocytes from survivors would be used to empirically identify antigens that elicit immune responses that strongly correlate with protection.

Multicistronic cassettes encoding ASFV antigens were successfully cloned into the pAd4 and pAd6 plasmid vectors and validated by sequencing. It was noted that cells transfected with the plasmid constructs expressed the target antigens, assembled virus, but the progeny virions were unstable. Concurrently, the collaborator who developed the Ad6 platform (Dr. M. Barry: Mayo Clinic) attempted to generate recombinant Ad6 viruses, but only managed to generate one stable virus construct encoding the p72-p15-B602L cassette [hereby designated SCAD24]. The virus was shown to express the ASFV antigens in HEK293A cells and in primary swine cells [Fig. 1], and authenticity of the expressed antigens was validated using lymphocytes from immunized pigs [Fig. 2]. Efforts to generate Ad4 virus constructs expressing ASFV antigens were not successful, but the technical challenges have now been solved (see section 2.10).

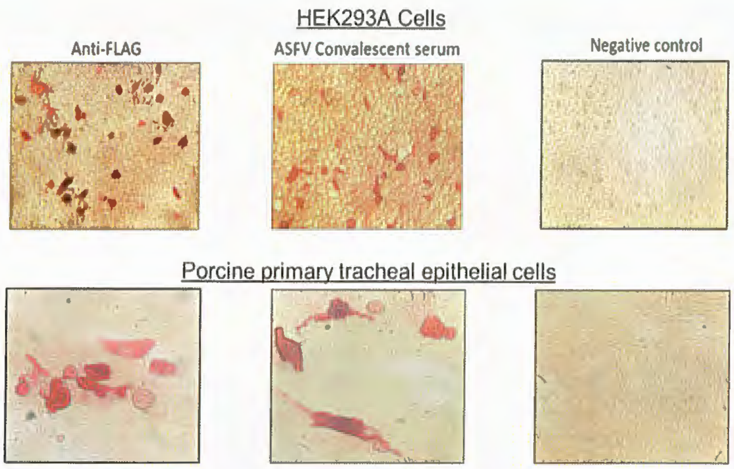

Figure 1. Protein expression by the SCAD24 virus. HEK293A cells [upper panel] or primary swine tracheal epithelial cells [lower panel] were infected with 10^6 SCAD24 and the cell monolayers were probed with anti-FLAG-AP or ASFV convalescent serum [followed by secondary anti-pig IgG-AP] 48 hrs. later. Expression was detected using Fast-Red substrate. Uninfected cells served as negative controls. The SCAD24 replicated efficiently in the primary swine cells and had extensive cytopathic effect.

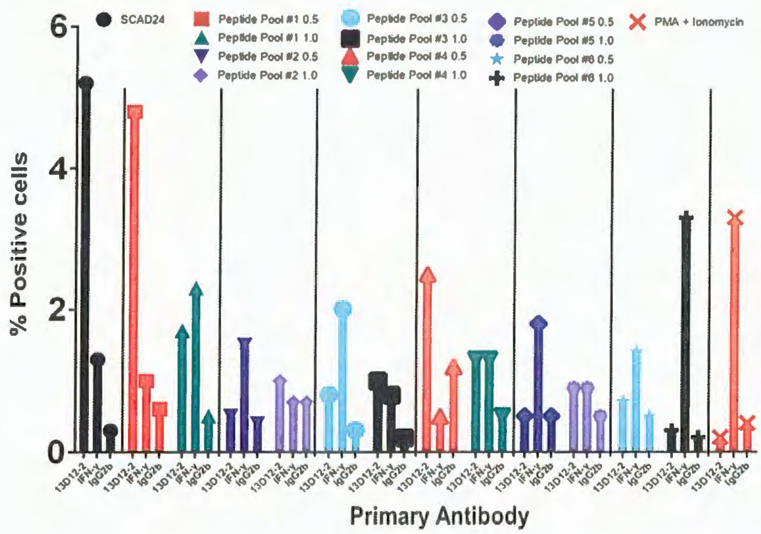

Figure 2. Validation of SCAD24. Lymphocytes from immunized pigs were used to confirm authenticity of the ASFV antigens expressed by SCAD24 using flow cytometry to enumerate porcine granzyme B^+ and IFN-γ^+ T cell responses following *in vitro* restimulation and intracellular staining. 13D12-2 is a porcine granzyme B-specific monoclonal antibody. Isotype control was a mouse IgG2b mAb. The peptides were synthesized using predicted *SLA* I binding motifs were included in the assay. Pools of 20 peptides were evaluated at different concentrations for their ability to stimulate granzyme B and IFN-γ responses.

In view of the above failure to generate Ad4 and Ad6 virus constructs, we opted to conduct immunogenicity and efficacy studies in pigs using recombinant replication-defective Ad5 (RD-Ad5-ASFV) virus constructs [SCAD24 was included] since these recombinant viruses are stable and express the encoded transgenes [Fig. 3].

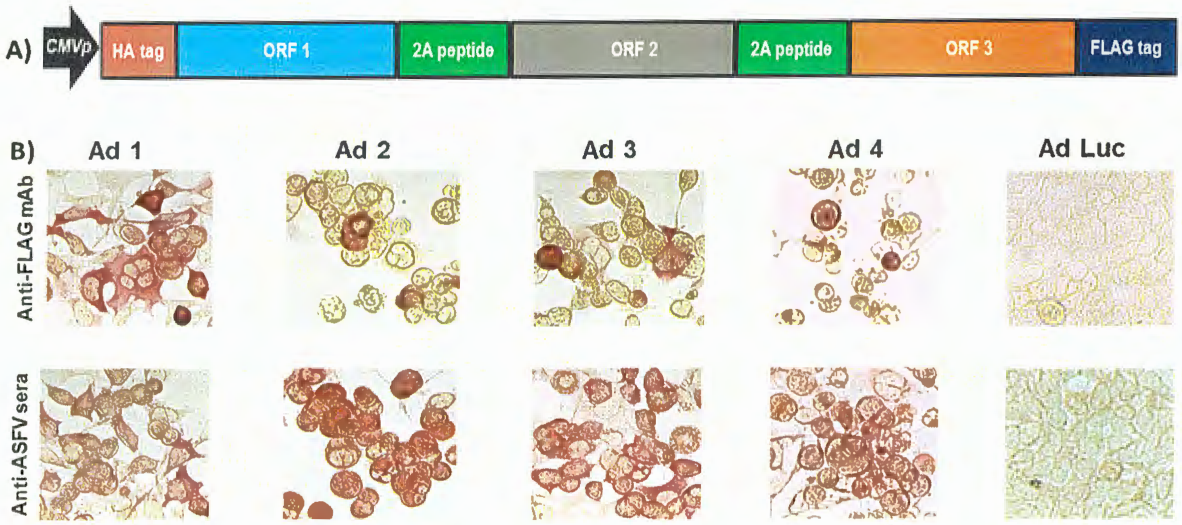

Figure 3. Design of multicistronic cassettes and validation of protein expression: **A**) Organization of the multicistronic expression cassettes for the ASFV antigens. Up to seven ORFs were included in a single cassette since the adenovirus vector can accommodate insert size of up to 8 kb. Codon-optimized synthetic genes were first cloned into pCDNA3 mammalian expression vector which has a strong CMV promoter [*CMVp*] and anti-HA/FLAG mAbs were used to confirm protein expression [data not shown]. **B**) Protein expression by recombinant adenoviruses [designated Ad 1 – Ad 4] was confirmed by immunocytometric analysis of infected cells using anti-FLAG mAb (upper panel) and more importantly, antigen authenticity was validated by the ASFV convalescent serum.

Safety, tolerability and efficacy of experimental subunit vaccines that were formulated, with and without adjuvant, using a cocktail of forty-two RD-Ad5-ASFV virus constructs, was evaluated in age-matched healthy commercial piglets. Since protective ASFV antigens are yet to be identified, immunization of pigs with a cocktail of the multicistronic expression cassettes was a strategy to deliver nearly 100% of the ASFV Georgia 2007/1 proteome to mimic antigen delivery by some ASFV mutants that have been shown to confer protection and if successful, this approach could fast track identification of the protective proteins needed for the development of rationally designed prototype subunit vaccines. The multicistronic expression cassettes were stable after multiple passages of the recombinant adenoviruses as judged by expression of the FLAG tag at the C-terminal of the last gene in the cassette. More importantly, the proteins expressed by the recombinant adenoviruses were shown to be authentic ASFV antigens as judged by staining using ASFV convalescent serum (Fig. 3 lower panel). However, *in vitro* expression of the ASFV antigens by the recombinant adenoviruses was heterogenous ranging from low (e.g. RD-Ad5-05, RD-Ad5-13), medium (e.g. RD-Ad5-18, RD-Ad5-39), to high (e.g. RD-Ad5-09, RD-Ad5-19) expression (Table 1). This outcome could have been influenced by the gene combination or arrangement in each cassette, but it was noted that protein expression by the majority of the cassettes was comparable or better than the positive control recombinant adenovirus encoding the pp62 antigen alone (Table 1). It is possible that the heterogenous expression of the ASFV antigens could have had an impact on the magnitude of priming B cell and T cell responses, but this study did not ascertain whether this was reflected in the resultant immune responses.

	Construct	% Positive cells
	None (negative control)	0.07
	Ad5-pp62 (positive control)	2.97
Low ASFV antigen Expression	Ad5-05	0.53
	Ad5-13	0.75
	Ad5-34	1.53
	Ad5-03	1.71
	Ad5-01	1.92
Medium ASFV antigen Expression	Ad5-04	2.38
	Ad5-29	2.87
	Ad5-11	3.08
	Ad5-10	3.14
	Ad5-18	3.29
	Ad5-39	3.44
High ASFV antigen Expression	Ad5-21	4.55
	Ad5-09	7.78
	Ad5-19	8.92

Table 1. ASFV antigen expression. Protein expression by HEK293A cells infected with the RD-Ad5-ASFV virus constructs was evaluated by flow cytometry analyses following intracellular staining using ASFV convalescent serum. The Ad5-pp62 served as the positive control (previously validated) and the RD-Ad-luciferase-infected cells served as the negative control. Values are expressed as a percent of the total parental cell population. Protein expression by representative RD-Ad5-ASFV constructs is presented.

Homologous prime-boost immunization of piglets with three doses of a cocktail of the recombinant RD-Ad5-ASFV virus constructs expressing the ASFV antigens or the RD-Ad5-Luciferase negative control construct, formulated with or without adjuvant, were well tolerated since all the piglets remained healthy throughout the immunization phase (Table 2 and Fig. 4). Notably, the priming dose of the RD-Ad5-ASFV cocktail alone or formulated with the Montanide ISA-201™ adjuvant (ASFV-ISA-201), but not the RD-Ad5-ASFV cocktail formulated in BioMize® adjuvant (ASFV-BioMize), elicited significant ($p<0.0001$) pp62-specific IgG responses compared to the RD-Ad5-Luciferase negative control immunogen (Fig. 5B). Boosting significantly ($p<0.0001$) recalled pp62-specific IgG responses in the vaccinees, but there was no difference in pp62-specific IgG responses between the pigs immunized with the RD-Ad5-ASFV cocktail alone and the pigs immunized with either the RD-Ad5-ASFV ISA-201 or the RD-Ad5-ASFV BioMize® formulations, suggesting that the adjuvants were not necessary (Figure 5C). This outcome does not imply that the adjuvants did not have an effect on the antibody isotype or T cell responses and/or functions of the induced effectors since these were not evaluated in this study. More importantly, the induced antibodies strongly recognized ASFV-infected, but not uninfected, primary swine cells (Figure 6). This outcome confirmed that the RD-Ad5-ASFV immunogens generated using synthetic genes induced authentic IgG responses.

Table 2: Immunization protocol.

Groups	Pig ID	Dose per Construct	Total Dose per Pig	Adjuvant
CS* Group 1	6908 6886	none		
Luc-ISA-201	6894 6909 6896 6897 6888	4.2 x 10^{11} ifu	4.2 x 10^{11} ifu	Montanide ISA-201™
CS* Group 2	6915 6901	none		
ASFV-No Adjuvant	6898 6903 6893 6892 6900	1 x 10^{10} ifu	4.2 x 10^{11} ifu	No Adjuvant
CS* Group 3	6887 6906	none		
ASFV-ISA-201	6884 6885 6883 6902 6910	1 x 10^{10} ifu	4.2 x 10^{11} ifu	Montanide ISA-201™
CS* Group 4	6912 6907	none		
ASFV-BioMize	6913 6891 6890 6899 6904	1 x 10^{10} ifu	4.2 x 10^{11} ifu	BioMize®

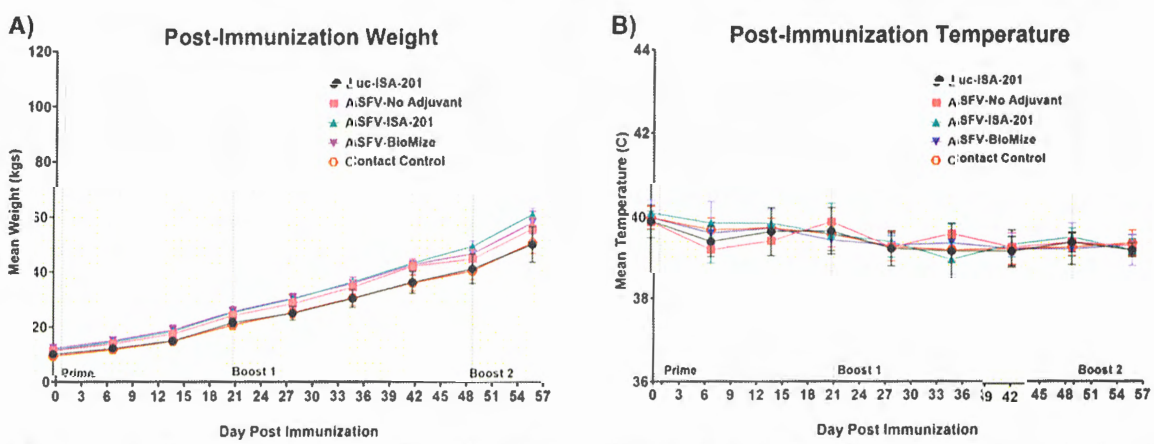

Figure 4. Post-immunization body temperatures and weights. (**A**) Weekly weight gain and (**B**) temperature change post-immunization. Average weekly weight gain and temperature change as clinical indicators of immunization effects for each group were plotted as group averages. Prime-boost immunizations are indicated on the days of administration. Contact Spreaders were not immunized and are indicated with a red open octagon.

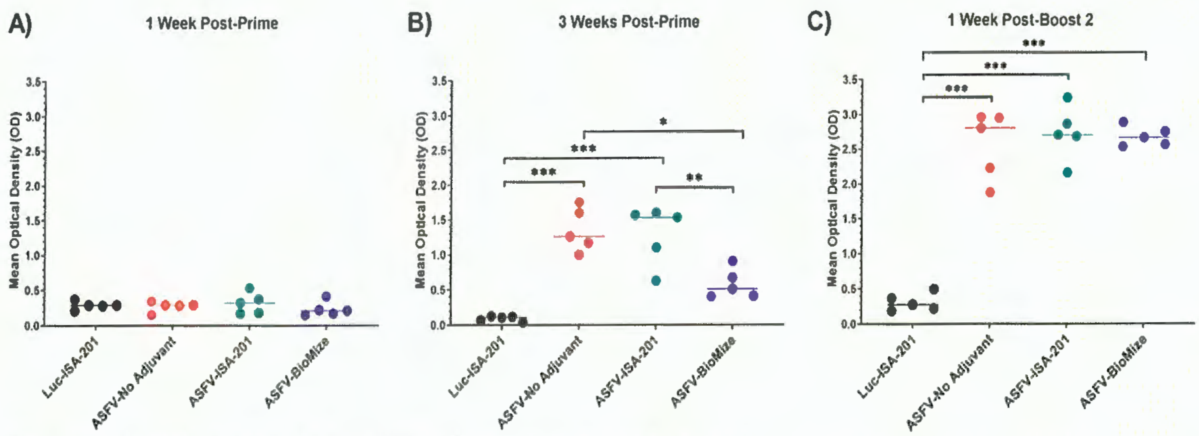

Figure 5. RD-Ad5-ASFV elicited antibody responses. Following priming and boosting, recombinant pp62 antigen was used to track IgG responses by ELISA using sera from blood collected: (**A**) one-week post-priming; (**B**) three weeks post-priming; or (**C**) one week after the second boost. Mean responses for the treatment groups are denoted by bars and statistically significant differences between the groups are denoted by asterisks: *p=0.002; **p=0.005; ***p< 0.0001, any groups not denoted with an asterisk were not statistically significant.

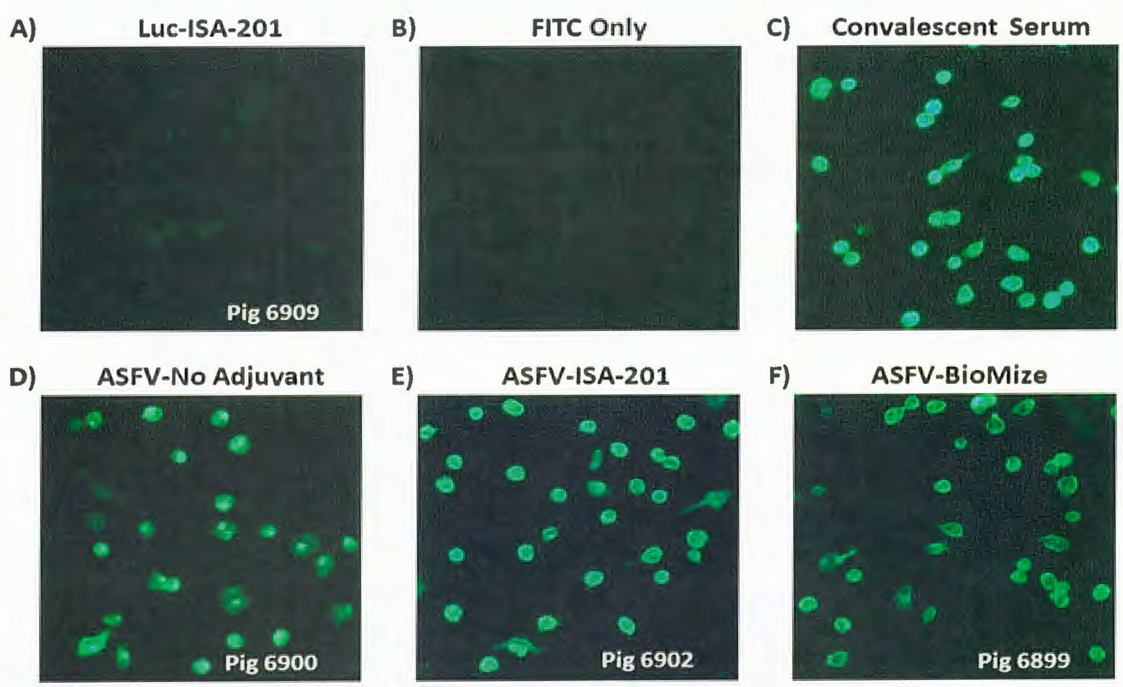

Figure 6. Antibodies elicited by RD-Ad5-ASFV recognized ASFV-infected cells. Antibodies induced by the RD-Ad5-ASFV formulations recognized ASFV-infected swine PBMCs as judged by IFA using sera from blood collected one-week after the second boost. Data for representative pigs from the negative control and the vaccinees is shown.

Following challenge of the pigs by exposure to comingled naïve contact spreaders that had received IM injection of ASFV (Georgia 2007/1), all the negative controls and the vaccinees, with one exemption, succumbed to ASF within fourteen days (Figs. 7 and 8). The contact challenge is a more realistic model for natural infection since ASFV transmission by ticks is limited to Sub-Saharan Africa, while in other regions of the world, the spread of virus to naïve pigs is by ingestion of contaminated materials or exposure to infected pigs. In this natural ASFV transmission model, the infection may occur via the mucosal route through direct animal to animal contact or oral-fecal route, likely with multiple low-dose repeated exposures more representative of non-tick ASFV transmission. One week after the initiation of the challenge, the pigs immunized with the RD-Ad5-ASFV cocktail formulated with or without adjuvant, lost weight, developed high fever and viremia in blood as well as in nasal fluids (Figures 7, 8A and B). Viremia in blood was significantly lower in the pigs immunized with the RD-Ad5-ASFV cocktail formulated in adjuvant compared to the naïve contact spreaders (Figure 8A). The pigs were terminated by day fourteen after they developed clinical ASF (Figure 8C). There was no significant difference in clinical presentation of ASF at termination between the naïve contact spreaders and the treatment groups. The RD-Ad5-Luciferase controls with an average clinical score and the survivor (immunized with the RD-Ad5-ASFV cocktail without adjuvant) which had no clinical symptoms at termination, were the only exemptions (Fig. 8C). The low clinical score noted for the RD-Ad5-Luciferase controls could have been due to the presentation of peracute ASF where animals are viremic but can show little to no outward signs of disease.

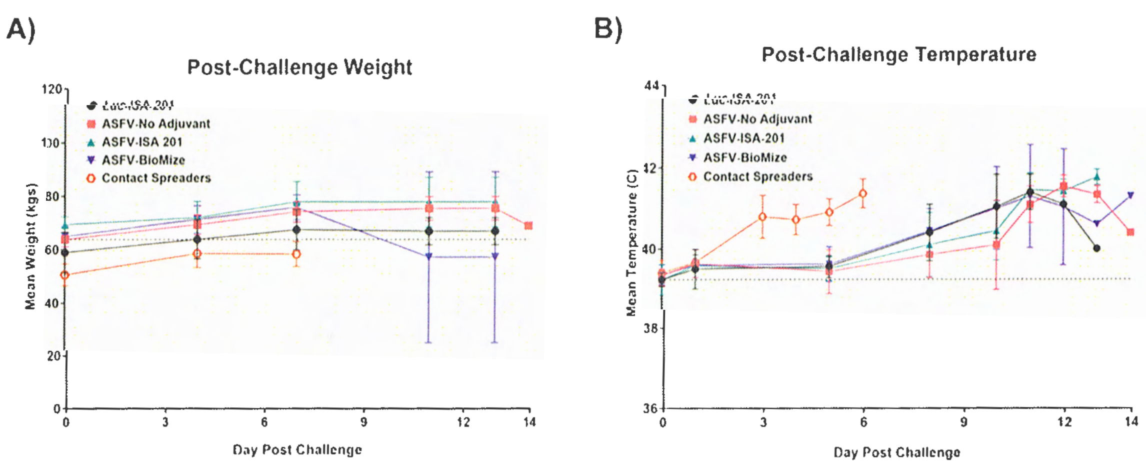

Figure 7. Post-challenge body temperatures and weights. Post-challenge weights (**A**) and temperatures (**B**) were collected bi-weekly. Contact Spreaders are indicated with a red open octagon. ASFV-No Adjuvant 14 DPC weight indicates the surviving animal only. Parameters were monitored as a clinical indicator of ASFV infection and disease progression for each group.

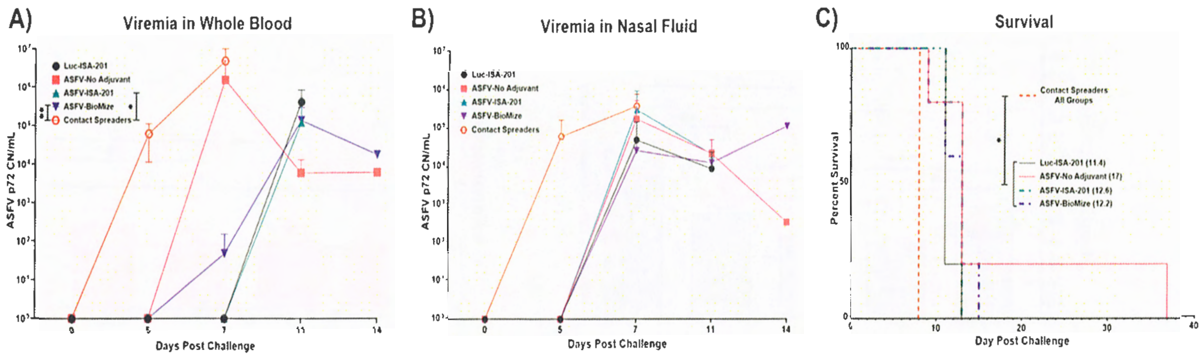

Figure 8. Survival and viremia post-challenge. Viremia in (**A**) blood and (**B**) nasal swabs from contact spreaders on days 0, 5, and 7; and blood from the treatment groups on days 0, 5, 7, 11, and 14 which includes the day of euthanasia for the animals that developed severe clinical disease. The mean viremia was significantly different between the contact spreaders and Montanide ISA-201™ (*p=0.417) and BioMize (**p=0.0439) treatment groups, any groups not denoted with an asterisk were not statistically significant and the survivor is not represented here. **C**) Pigs were monitored and euthanized based on disease severity. All the non-immunized contact spreaders (CS) are shown in red with dashed lines. Average survival is indicated in parenthesis for each group.

Histopathology of vaccinees revealed lesions that were consistent with moderate subacute ASF which were consistent with the observations made from the naïve spreaders as well as the RD-Ad5-Luciferase negative controls (Fig. 9). At termination, the mean viremia in tissues from the naïve contact spreaders was significantly ($p<0.0001$) higher compared to the mean viremia in tissues from the RD-Ad5-Luciferase negative controls and the pigs immunized with the RD-Ad5-ASFV cocktail alone or formulated in adjuvant (Figure 10). However, there was no significant difference in tissue viral load in the RD-Ad5-ASFV vaccinees when compared to the RD-Ad5-Luciferase negative controls (Figure 10). In contrast, it was observed that the mean histopathological scores of tissues from the contact spreaders were higher than the mean scores for the RD-Ad5-Luciferase controls and the RD-Ad5-ASFV vaccinees. However, at the individual level, there was wide variation in total tissue histopathological scores between the treatment groups and among the individual control and vaccinated pigs. Overall, the outcome showed that immunization of pigs with the experimental RD-Ad5-ASFV formulation was not protective even though the immunogens induced antibody responses that recognized ASFV-infected cells. However, in this study, only anti-pp62 antibody responses were evaluated to track seroconversion and therefore, it is not known whether the vaccinees mounted antibodies against the other antigens included in the cocktail. Although ASFV neutralization has been reported, the role of antibodies in protection against ASFV infection is yet to be determined.

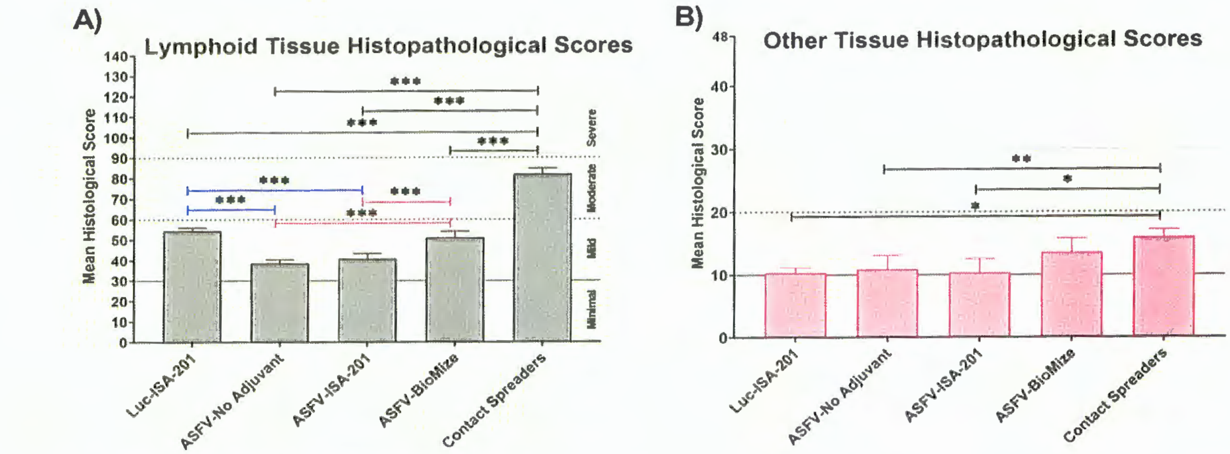

Figure 9. Tissue histopathology. Mean histological scores per group for (**A**) lymphoid tissue and (**B**) other tissues (liver, lungs, kidney, heart). Group comparisons are indicated by *p=0.0014, **p=0.0057, ***p<0.0001, and any groups not denoted with an asterisk were not statistically significant.

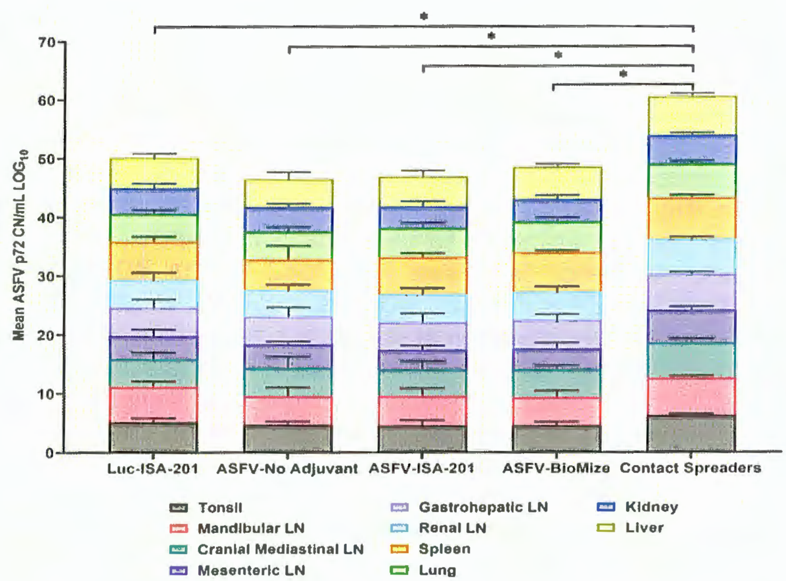

Figure 10. Viremia in infected tissue. Tissue CN/mL was quantified by qPCR from the samples collected on the day of euthanasia for the vaccinees and Contact Spreaders. Respective tissue types are denoted by separate colors. Each plotted value is stacked and represents the mean for each tissue. The Log base 10 was calculated for each mean CN/mL and plotted in a stacked bar graph. Total mean ASFV p72 genomic DNA was statistically significant different between the contact spreaders and all other groups denoted by asterisks, *p< 0.0001, any groups not denoted with an asterisk were not statistically significant.

One pig (number 6892) immunized with the RD-Ad5-ASFV cocktail without adjuvant was the only survivor (Figure 8C). Although the pig mounted a rapid increase in pp62-specific IgG responses after priming, there was no significant increase in antibody response after boosting and it had the lowest antibody responses compared to the other four pigs in this group (Figure 11A). It was also noted that viremia peaked one week after initiation of contact challenge, followed by subtle weight loss and high fever that peaked on the twelfth day (Figure 11B). By day fourteen when all the other pigs in this group were terminated, the pig started to gain weight and sustained it for the next twenty-five days during which time the fever resolved and viremia in blood gradually decreased, but there were recurrent episodes of high viremia in nasal fluid until study termination (Figures 8C and 11B). The viral detection in nasal fluid likely reflects transient reinfection from the environment for this pig, however the environmental viral loads were not monitored in this study to confirm this possibility. The pig did not exhibit typical ASF clinical symptoms, but necropsy revealed that it had chronic pericarditis and lesions consistent with moderate subacute ASF. At termination, tissue samples had the lowest amount of ASFV compared to the average viremia detected in the contact spreaders as well as the pigs from all the other groups (Figures 11C). The recurring virus shedding, low viremia in tissues, and the lesions observed were consistent with outcomes observed in pigs with chronic ASF. Although this pig had the lowest anti-pp62 antibodies, B cell responses against other antigens included in the RD-Ad5-ASFV cocktail were not determined and in addition, T cell responses were not evaluated. A previous study has shown that, following immunization, vaccinees that mounted the lowest, but not the highest, antigen-specific IgG responses had better survival rate and lower clinical scores. It is possible that the survivors mounted low antibody responses that were directed against protective determinants, whereas the non-survivors had strong responses against non-protective antigens. It is also likely that the sole survivor had unique genetic traits that enabled it to resolve the virus which would be consistent with the observation that even infection of naïve pigs with the most virulent ASFV isolates does not always result in 100% mortality.

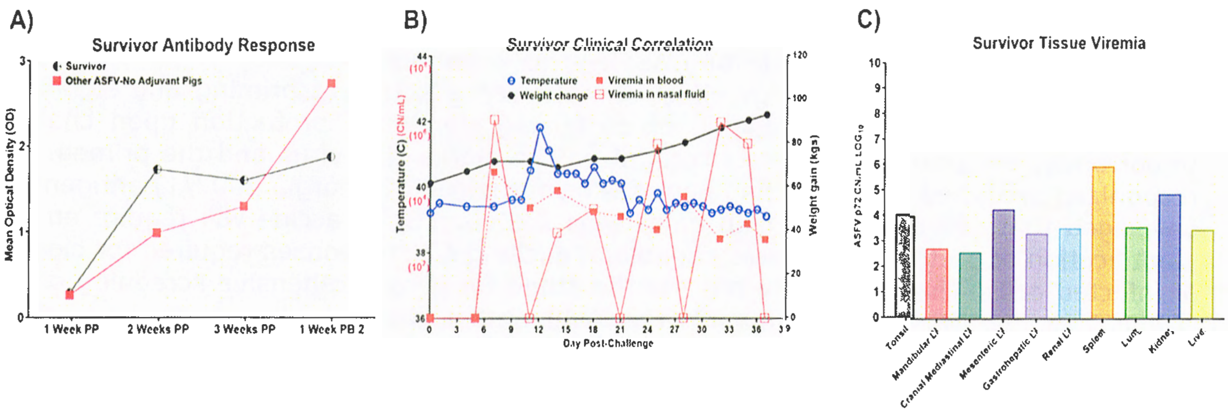

Figure 11. Ad5-ASFV-vaccinee that survived. A) Antibody response by the lone survivor that was immunized with the RD-Ad5-ASFV without adjuvant. The half-circle indicates the pp62-specific IgG responses by the survivor compared to the mean IgG responses (solid square) by the other four group mates that succumbed after challenge. Post-prime is indicated by PP and post-boost by PB. **B**) Temperature, weight change, and viremia in blood and nasal fluid post-challenge. **C**) Viremia expressed in Log base 10 of the CN/mL for each respective tissue.

The results from this study showed that immunization of pigs with a cocktail of forty-two multicistronic ASFV antigen expression constructs, formulated with and without adjuvant, primed humoral immune responses as judged by anti-pp62 IgGs, which underwent significant recall after boosting. However, none of the RD-Ad5-ASFV formulation conferred protection upon challenge except one sole survivor that had been immunized with the RD-Ad5-ASFV cocktail without adjuvant. Even though the survivor had no typical clinical symptoms, it had viral loads and lesions consistent with chronic ASF. Given the near 100% coverage of the ASFV proteome, the outcome from this study suggests that *in vivo* antigen expression, but not the antigen content, could be the limitation of this immunization approach. This could be due, in part, to the fact that the replication-incompetent adenovirus does not mimic replication of attenuated ASFV since it does not amplify and persist in pigs to effectively prime and expand protective immunity. Additionally, the RD-Ad5 vector and live attenuated ASF virus do not use identical promoters or processes for gene expression, with attenuated ASFV genes being differentially transcribed ranging from early to late expression while the RD-Ad5 vectored multicistronic genes rely on a single CMV promoter for simultaneous co-expression. Protein expression by the RD-Ad5-ASFV constructs was heterogeneous and this outcome could have been influenced by the gene combination or sequence arrangement in each cassette. The heterogenous antigen expression could have had an impact on the magnitude of priming B cell and T cell responses, but this study did not ascertain whether this was reflected in the resultant immune responses. Addressing the limitations of the *in vivo* ASFV antigen delivery will likely yield promising outcomes. Based on our previous studies, the homologous prime-boost immunization regimen using the replication-incompetent Ad5 is well suited for induction and robust expansion of B cell, but not T cell responses. Future approaches will evaluate protective efficacy of replication-competent vaccine vectors expressing the multicistronic ASFV cassettes as well as heterologous prime-boost immunization. Other limitations of this study include the use of small sample size (n=5), but addressing this limitation is still a challenge since ABSL-3Ag biocontainment space required for ASFV challenge has limited capacity.

The data generated from this proof-of-concept (POC) study shows that the adenovirus vectored ASFV multicistronic expression cassettes and the immunization protocol that we have developed is capable of eliciting ASFV-specific immune responses. The study used a homologous prime-boost strategy which is not very effective at priming and significantly amplifying immune memory capable of conferring significant protection upon challenge. Importantly, the primed antibodies recognized the actual ASF virus and the primed T cells recognized predicted *SLA*-1 binding peptides from the ASFV (Georgia 2007/1) antigens. It is envisaged that, development of an efficacious ASFV subunit vaccine will require empirical identification of protective antigens capable of eliciting CTL responses required for clearance of infected cells. This POC study has set the stage for a comprehensive screening of ASFV antigens to identify such determinants for development of a subunit vaccine. In a separate project, the above-mentioned outcomes were used to generate and test safety, immunogenicity, and protective efficacy of the ASFV multicistronic expression cassettes delivered using a replicating antigen delivery vector and we have achieved promising protective efficacy [data from the project was used to generate the proposal submitted to FFAR, which is currently under review].

One of the major challenges in identification of protective ASFV antigens is lack of a informative readout platform to identify immune responses that correlate with protection, which would guide identification of the cognate antigens for subunit vaccine development. Emerging evidence suggest that cytotoxic T lymphocytes (CTLs) are essential for clearance of ASFV-infected swine cells. Thus, it is rational to identify novel CTL target antigens for inclusion in the next generation replicon-based prototype vaccines. We can induce ASFV antigen-specific CTLs, but validation entails enumeration of lytic activity using the standard ^{51}Cr release assays which are not easily done in BSL-3 biocontainment. The practical approach is intracellular staining [ICS] and flow cytometric enumeration of granzyme-B/perforin expression since this are the key effector molecules utilized by CTLs and NK cells for target lysis. However, a monoclonal antibody (mAb) against swine granzyme-B is not available. We have generated mAbs against swine granzyme-B to enable us to identify ASFV antigens that elicit porcine granzyme-B expression [Fig. 12]. One mAb [13D12-2] is working well by intracellular staining (ICS), whereas mAb 14G9-2 is an excellent probe for immunohistochemistry [Fig. 12]. The mAb 13D12-2 was used to identify seven ASFV antigens [A118R, 173R, M1249L, G1211R, A859L, B318L, B169L] by using predicted *SLA*-I-binding peptides that recalled strong ASFV-specific granzyme-B^{+}IFN-γ^{+}CD8α^{+} T cell responses (Fig. 13A). Fine mapping revealed that the A859L antigen contains two epitopes (STDETRTAI; and LVDQGVYAL) that induced strong granzyme-B^{+}IFN-γ^{+}CD8α^{+} T cell responses (13B). Importantly, the epitopes are 100% conserved among all the sequenced ASFV genomes and they are restricted by multiple *SLA-I* alleles. Therefore, the A859 antigen is an attractive candidate to be considered for development of a broadly protective ASFV subunit vaccine.

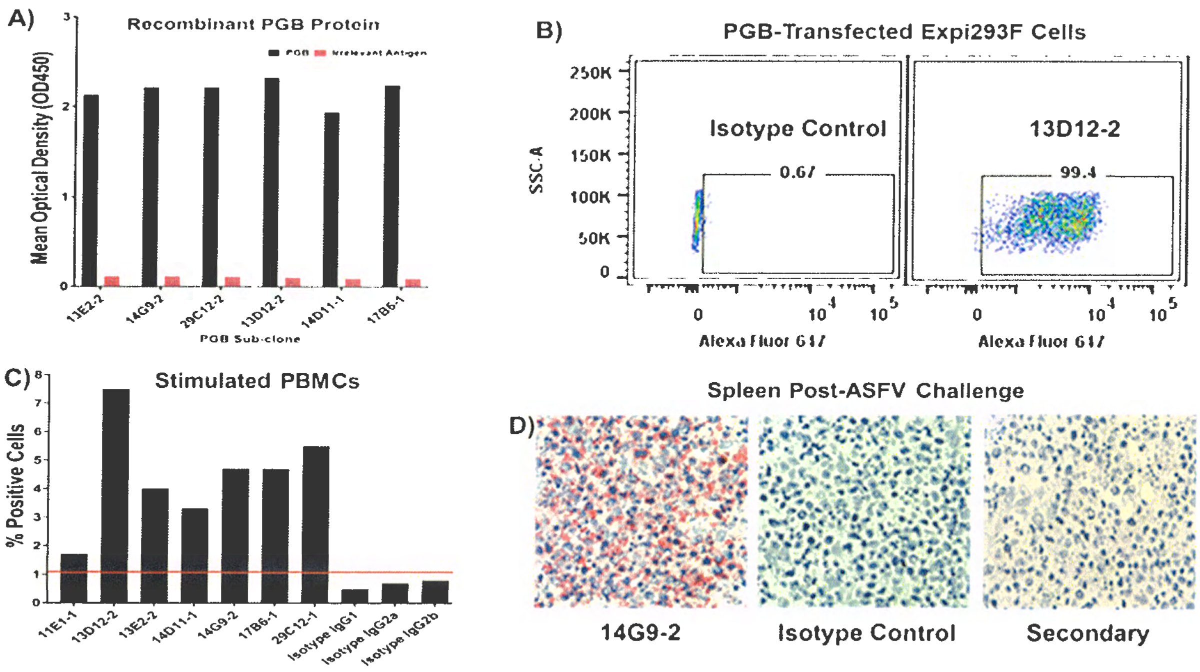

Figure 12: **Characterization of mAbs against porcine granzyme-B**: Candidate mAbs were screened: **A**) by ELISA using recombinant porcine granzyme-B (PGB); **B**) by ICS of cells transfected with the gene encoding porcine granzyme-B and probed with IgG isotype control or mAb 13D12-2; **C**) ICS of PBMCs isolated from an immunized pig and then stimulated *in vitro* for seven days with recombinant Ad6 virus co-expressing p72-p15-B602L [the pig was immunization with a cocktail of recombinant Ad5 viruses expressing ASFV antigens that contained RD-Ad5-p72-p15-B602L]. Results above the redline were considered positive. PBMCs from sham immunized pig [RD-Ad5-Luciferase] were below the 1% level [data not shown]; and **D**) Formalin-fixed paraffin-embedded porcine spleen tissue collected at necropsy post-ASFV challenge was probed with mAb 14G9-2 or IgG isotype control, whereas the secondary antibody alone served as control for non-specific binding. Several mAbs for ICS and IHC were identified and representative data is shown [*Zajac, M., et al., 2023: manuscript in-preparation]*.

A)

Antigens M1249L, A118R, 173R		Antigen G1211R		Antigens A859L, B318L, B169L	
Pool A		Pool B		Pool C	
Peptide ID	Sequence	Peptide ID	Sequence	Peptide ID	Sequence
1	ASLEFNTFY	19	AADDTTCYY	33	AIDIHQGGW
2	AVDSAVRIF	20	IAYAIHDEY	34	ALFHGIHPL
3	KADSSLNAY	21	ITYLNNMGY	35	FQNPFIVAL
4	KIDKTNTIY	22	KIDDIIGDY	36	FTDEWKYCF
5	LLEKMMQQY	20	KLDTILKCY	37	MIERAAHCF
6	NIIQFTHQY	23	KVDLPYHLM	38	YIEKYWPGY
7	QPELVHDYY	24	MVDCCHKNY	39	YLVPGLSLL
8	RAYELFLKY	25	RIENIPVFF	40	YPWEESMAF
9	RLIDFLFTL	26	RRMLYNEPL	41	YTLCDSPAY
10	RLMAVDSAV	27	ALEADNKNY	42	IVSSIHDKY
11	SAYSSPGLF	28	GILSYTSLY	43	LLNPSEVAY
12	SSMDDISAY	29	IGDYPDTFY	44	QLARHAIRY
13	STMYSVSPV	30	PIDIHEVRY	45	STDETRTAI
14	VLDNLAPQM	31	YITCPDSLY	46	LVDQGVYAL
15	VLESIRVLY	32	YLQKFVQSI		
16	VTDPASALL				
17	VVNEYHTTF				
18	YAYILNLIY				

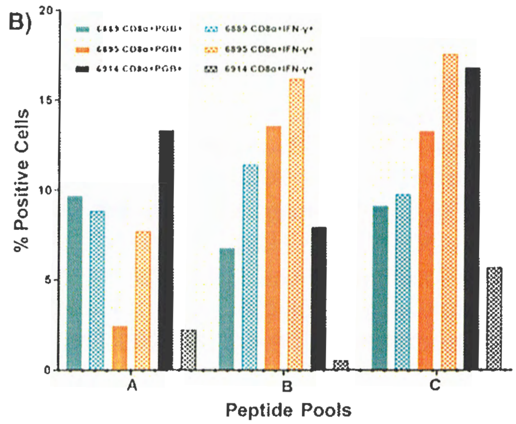

Figure 13. Predicted *SLA-I* binding peptides elicit recall responses. Representative data from assays conducted to validate putative CTLs epitopes are shown: **A**) Synthetic peptides based on predicted strong *SLA*-I binding motifs from the ASFV Georgia 2007/1 antigens (A118R, 173R, M1249L, G1211R, A859L, B318L, B169L) were tested for their ability to stimulate recall responses of PBMCs from immunized pigs; **B**) PBMCs [from pigs: 6889-teal; 6895-orange; 6914-black] were stimulated *in vitro* for five days with low dose [1 MOI] of ASFV and then used to screen peptide pools [pools A, B, and C are shown] for granzyme-B$^+$ and IFN-γ^+ expression by CD8α^+ T cells using flow cytometry. Sham-vaccinees were negative. Positive control was a mitogen cocktail (PMA + Ionomycin), whereas irrelevant stimuli and media alone served as negative controls. Live CD8α^+granzyme-B$^+$ or CD8α^+IFN-γ^+ lymphocytes are reported as percent-positive cells after subtracting background from irrelevant stimulant. Fine mapping resulted in identification of potent granzyme-B$^+$ and IFN-γ^+ T-cell epitopes [example from antigen A859L: A859L156-164 (STDETRTAI); and A859L428-436 (LVDQGVYAL) [*Zajac, M., et al., 2023: manuscript in-preparation*].

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

Quality Assurance/Quality Control: Our lab has written Standard Operating Procedures (SOPs) that guide *in vitro* and *in vivo* studies. Data are acquired and recorded in lab books or archived on backed up hard disks. Authenticity of genes encoding vaccine candidates are routinely validated by DNA sequencing, protein expression is confirmed by immunocytometric analyses/IFA/ICS/ELISA/Western Blotting using validated convalescent polyclonal serum, antigen-specific monoclonal antibodies (mAbs), and /or mAbs against tags fused in-frame to the target antigens. T-cell subsets are validated by flow cytometry using commercial mAbs. In all assays, relevant controls (positive and negative) are included. T-cell assays are conducted in triplicates and repeated whenever necessary. In addition, cells from each assay time point are frozen for repeat assays if need be or for additional alternative readout assays. Statistical analyses are performed to determine the significance of the difference in readouts between the treatments and respective controls. Power analyses are used to determine minimum number of animals required for *in vivo* studies. Results are discussed in lab meetings and reviewed by the principal investigator for accuracy. Equipment in our labs are routinely cleaned and calibrated, pipettes are calibrated every year, and inspected for accuracy. Our labs are inspected annually and we have approval to conduct studies at BSL-2/ABSL-2Ag/BSL-3/ABSL-3Ag containment. We have pertinent institutional/state/federal permits to work with pathogens. The research personnel have received requisite training.

2.8. List key stakeholders that could be served by results of this project.

Pork industry stakeholders including small- and large-scale producers, processors, suppliers of swine industry inputs, consumers of pork and pork products, commercial entities that are interested in the development of a safe and efficacious ASFV vaccines, federal and state policy makers, FFAR/USDA/DHS/NPB/SHIC, NIFA project directors and researchers in ASFV and other swine diseases, WHOA/FAO, and international entities (e.g. European Union) focused on the risks posed by ASFV to the swine industry.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sourceswill be reported under Information Products. (up to 1,000-word limit)

Nothing to report.

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

The overarching goal of this study was to utilize two viral replicon vectors [Adenovirus 4 (Ad4) and Adenovirus 6 (Ad6)] to develop and evaluate immunogenicity as well as protective efficacy of two next generation prototype subunit vaccines, designated Ad4-PV and Ad6-PV, encoding rationally selected multiple ASFV antigens containing putative Cytotoxic T lymphocytes (CTLs) epitopes. Recombinant Ad4 constructs encoding ASFV antigens expressed the transgenes, however, we were unable to generate viable recombinant Ad4-ASFV viruses that could be amplified due to instability. The ASFV genes were successfully subcloned into the pAd6 shuttle and Ad6 destination vectors and shown to express ASFV antigens [Data not shown]. Sequencing validated the authenticity of the genes in the destination vectors. Recombinant constructs generated expressed transgenes, but we were unable to successfully assemble stable recombinant Ad6 viruses even after trying two destination vectors and a new cell line (Bleo 10 cells) developed by Dr. Barry [vector developer: Mayo clinic]. It was noted that the Bleo 10 cells transfected with the Ad6-ASFV plasmid constructs expressed the target antigens, assembled virus, but the progeny virions were unstable. Concurrently, the Barry lab. attempted to generate recombinant Ad6 viruses, but only managed to generate one stable virus construct encoding the p72-p15-B602L cassette [hereby designated SCAD24]. The virus was shown to express the ASFV antigens in HEK293A cells and in primary swine cells, and the antigens were recognized by lymphocytes from immunized pigs. Immunization of pigs with a formulation containing SCAD-24 revealed that the pigs responded, but the virus was shed and it caused diarrhea.

Due to the need for a heterologous prime-boost immunization strategy to improve induction and expansion of ASFV antigen-specific immune responses *in vivo*, we sought help from a NIH collaborator [Dr. Mark Connors: Ad4 vaccine vector expert] and after receiving hands-on training in his lab using a new generation vector platform, we are now assembling recombinant Ad4 viruses encoding ASFV antigens. Since Ad6 was shed by pigs and caused diarrhea, we pursued an alternative approach by generating, in-house, attenuated replication-competent adenovirus-5-vectored constructs and the resulting recombinant viruses expressing ASFV antigens are stable.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

One undergraduate student, eight DVM students, three PhD students, and one postdoc received hands-on training. These individuals took pertinent CITI training courses required for federal and institutional research compliance. Specifically, the undergraduate student received training on preparation of reagents and conducting *in vitro* experiments at BSL-2 level. The DVM students received hands-on training on monitoring health status, handling, and bleeding pigs. They also participated in pig immunizations. Two graduate students and the postdoc received hands-on training in generation and purification of adenovirus-vectored ASFV antigen expression constructs as well as quality control validation. All the graduate students and the postdoc received hands-on training in pig immunization, bleeding and processing blood for sera and peripheral blood mononuclear cells. The DVM students, the graduate students, and the postdoc received training on how to work with pigs in ABSL-2Ag biocontainment, and in addition, the graduate students, and the postdoc received training on how to work with ASFV-infected pigs in ABSL-3Ag biocontainment. Three graduate students were trained on determination of viremia in blood and tissue samples using qRT-PCR as well as conducting immune response readouts at BSL-3 containment by flow cytometry/IFA/ELISA. All the graduate students and the postdoc were trained on humane animal pig euthanasia in ABSL-3Ag biocontainment, and necropsy and tissue processing under guidance of a pathologist. One graduate student received in-depth training on histology and immunohistochemistry, while two students got trained on flow cytometric analyses and EliSpot assay. In addition, a Research Associate and Research Assistant Professor received hands-on training.

2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

One undergraduate student, eight DVM students, three PhD students, one postdoc and one research associate were involved in this study.

3. Information Products

3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

1. Zajac, M.D., J. Yao, R. Kumar, N. Sangewar, S. Lokhandwala, H. Sang, K. Mallen, J. McCall, L. Burton, D., Kumar, E. Heitmann, T. Burnum, S. D. Waghela, K. Almes, J. D. Trujillo, J. Richt, T. Kim, W. Mwangi. 2023. Immunization of pigs with replication-incompetent Adenovirus-vectored African Swine Fever Virus multiantigens induced humoral immune responses but no protection following contact challenge. *Front. Vet. Sci.* Volume 10 - 2023. https://doi.org/10.3389/fvets.2023.1208275.
2. Zajac, M.D., J. McCall, N. Sangewar, B. Libanori-Artiaga, L. Burton, D. Madden, E. Heitmann, J. Yao, R. Kumar, H. Sang, J. Richt, T. Kim, W. Mwangi. Identification of novel African Swine Fever Virus antigens that induce robust Granzyme-B^+IFN-γ^+CD8α^+ T cell response in pigs. 2023. Manuscript in-preparation.
3. Mwangi, W., R. Kumar, M. Zajac, H. Sang, S. Adetunji, J. Trujillo, J. Richt, J. Manuel-Vizcaino, and T. PARTNERSHIP: Single-cycle replicon-based African Swine Fever virus subunit vaccine. CRWAD 2023. Abstract # 77005.

3.2. Please provide a list of citations for the information products produced during theproject.

1. Zajac, M.D., J. Yao, R. Kumar, N. Sangewar, S. Lokhandwala, H. Sang, K. Mallen, J. McCall, L. Burton, D., Kumar, E. Heitmann, T. Burnum, S. D. Waghela, K. Almes, J. D. Trujillo, J. Richt, T. Kim, W. Mwangi. 2023. Immunization of pigs with replication-incompetent Adenovirus-vectored African Swine Fever Virus multiantigens induced humoral immune responses but no protection following contact challenge. *Front. Vet. Sci.* Volume 10 - 2023.

https://doi.org/10.3389/fvets.2023.1208275.

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3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Zajac, M.D., J. Yao, R. Kumar, N. Sangewar, S. Lokhandwala, H. Sang, K. Mallen, J. McCall, L. Burton, D., Kumar, E. Heitmann, T. Burnum, S. D. Waghela, K. Almes, J. D. Trujillo, J. Richt, T. Kim, W. Mwangi. 2023. Immunization of pigs with replication-incompetent Adenovirus-vectored African Swine Fever Virus multiantigens induced humoral immune responses but no protection following contact challenge. *Front. Vet. Sci*. Volume 10 - 2023. https://doi.org/10.3389/fvets.2023.1208275.

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

N/A

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s). No.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if theymust remain confidential indefinitely. (up to 500-word limit). N/A

3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers,and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access andre-use these data in the future? N/A

4. Data Management

4.1. Did the project generate any data? Data generation includes transformation ofexisting data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award. N/A

4.2. If you list multiple data sets, are these data sets related? If so, please provide ashort description of how they are related. N/A

4.3. Please provide copies of relevant metadata records to support FFAR's mission ofenhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder. N/A

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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I willnot claim otherwise.

PI Full Name: Waithaka Mwangi

PI Signature: Waithaka Mwangi Digitally signed by Waithaka Mwangi Date: 2023.10.25 15:13:26 -05'00' Date: 10/25/2023

Authorized Signing Official (ASO) Name: Mr. Paul Lowe

ASO Signature: Paul R. Lowe Date: 10/30/2023